Quantifying the sensitivity of a mesh-based tree segmentation pipeline to classifier precision, Bayesian refinement configuration, and stand structure

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Abstract

Terrestrial laser scanning (TLS) enables individual tree segmentation at plot scale, but the operational adoption of TLS-based forest inventory depends on automated pipelines that can reliably transform millions of raw points into labeled individual trees. We present a two-stage pipeline consisting of *TreeSegPy*, a KPConv-based point classifier that assigns each TLS point to the tree or non-tree class, and *TopTreeSegR*, a mesh-based topological segmenter that partitions the filtered point cloud into individual tree instances. *TreeSegPy* achieves a tree-class F1 of 0.90 ± 0.05 across five held-out conifer-dominated plots (three pine, two spruce–pine), with a consistent optimal probability threshold of 0.10 across all plots, so no per-plot calibration is required. Downstream, we identify two independent failure modes that limit the operational reliability of the mesh-based segmenter. First, the default configuration of *TopTreeSegR*'s Bayesian Boundary Refinement (BBR) stage, *confidence_threshold* = 1.0, increases the odds of catastrophic metric failure, and the effect is metric-specific: 7.9-fold [5.7, 10.9] for stem diameter, 3.5-fold [2.6, 4.8] for crown volume, and 1.8-fold [1.2, 2.7] for height. The effect is dose-dependent — higher thresholds and disabling BBR entirely reduce the failure rate toward the baseline — and is not mitigated by finer mesh resolution. Second, the baseline failure rate without BBR varies substantially across plots, from 4% on Milicz to 47% on Katrynka, and this variation is tightly predicted by the interquartile range of reference tree heights ($r = 0.96$, $R^2 = 0.93$, $n = 5$). The point classifier, despite its ARI cost on downstream segmentation, does not measurably increase the probability of catastrophic metric failure, indicating that the two failure modes are separable. We recommend disabling BBR by default or raising *confidence_threshold* to at least 1.5, and reporting the interquartile range of tree heights alongside segmentation results as a diagnostic for plots where baseline failures are expected to be high. A post-hoc per-plot DBH calibration is presented as a practical addendum with an explicit domain of validity, reducing the median absolute error on the typical tree from ~1.0 cm to ~0.2 cm on plots where the absolute-failure rate is low.

Keywords: Terrestrial laser scanning · Individual tree segmentation · Topological data analysis · Bayesian Boundary Refinement · Forest inventory · Failure analysis

1. Introduction

Terrestrial laser scanning (TLS) resolves forest structure at the level of individual trees. A single plot scan can yield hundreds of millions of points, from which tree height, stem diameter, crown dimensions, and stem density can in principle be derived directly (Calders et al., 2020; Liang et al., 2018). That promise has not translated into routine operational use. The obstacle is not acquisition — TLS instruments are portable and increasingly affordable — but processing: raw point clouds must be partitioned into individual tree instances before any inventory metric can be computed (Burt et al., 2019; Martin-Ducup et al., 2021). Individual tree segmentation (ITS) remains the rate-limiting step for TLS-based inventory.

A decade of ITS development has produced several method families, none of which has resolved the problem generally. Bottom-up methods detect stems at low height and grow crowns upward, but require parameter tuning that shifts with forest type, stem density, and point quality (Burt et al., 2019; Trochta et al., 2017). Top-down and deep learning approaches reach high accuracy on benchmark data but depend on large labeled training sets, are sensitive to domain shift, and are typically validated in forest types unlike the dense, multi-layered stands where segmentation is hardest (Chang et al., 2022; Krisanski et al., 2021; Peng et al., 2026; Wang et al., 2021; Xi and Hopkinson, 2021). Allometric methods embed ecological knowledge but are species- and site-specific. Recent benchmarking work has confirmed that no single method dominates across forest types (Cherlet et al., 2026).

Topological methods operate on the connectivity of the point set rather than its geometry alone. Xu et al. (2023) demonstrated a point-based topological segmentation using discrete Morse theory, evaluating it with point-level

instance-similarity measures — Rand index, normalized directional Hamming distance, and mean Intersection over Union — and showed that critical-point analysis can separate individual trees in TLS clouds. *TopTreeSegR* extends this principle from points to a mesh representation [CITE: Dioszegi et al., TopTreeSegR]. It builds an α-complex over a tree-only point cloud, computes a discrete gradient of the height function, and traces ascending manifolds from trunk-seeded minima to assign each mesh vertex to a tree. Operating on the mesh rather than on the raw points is what makes the method parameter-free at the user level and allows it to preserve the tree-level metrics that matter for inventory. It also defines the method's operational dependency: the mesh is constructed from tree points only, so a tree/non-tree classifier must precede it when the input is a raw TLS cloud. The α-complex is not robust to non-tree returns, which fragment the mesh or generate spurious components.

Tree/non-tree separation is therefore not a preprocessing detail but the upstream half of the pipeline. Chang et al. (2022) made this explicit by designing their two-stage TLS segmentation around a semantic network that first classifies points into tree and non-tree categories, precisely to remove understory vegetation that would otherwise degrade stem detection. Point-based deep learning architectures can separate woody from non-woody returns at high point-level accuracy (Krisanski et al., 2021; Thomas et al., 2019), but point-level accuracy is an incomplete description of what a classifier contributes downstream. A mesh-based segmenter responds to how classifier errors are distributed in space, not only to how many occur. Errors concentrated in the crown, where the mesh is locally well connected, may be harmless; errors concentrated in the stem region, where the trunk ring must remain intact for diameter estimation, may not be. Whether classifier precision measurably degrades a mesh-based segmenter's output — and whether it does so beyond its effect on aggregate accuracy — has not been tested. Nor has a tree/non-tree classifier been combined with a mesh-based topological segmenter in an end-to-end evaluation.

A second gap concerns the segmenter's own configuration. The evaluation of *TopTreeSegR* reports aggregate agreement between predicted and reference labels on the mesh, using ARI and F1 [CITE: Dioszegi et al., TopTreeSegR]. Aggregate accuracy averages over all mesh vertices and cannot distinguish a pipeline that is uniformly mediocre from one that is excellent on most trees and catastrophically wrong on a minority. For inventory, that distinction is decisive. A tree whose diameter is estimated at twice its true value is not an error to be averaged away; it is an unusable measurement that propagates into biomass and volume estimates. The tree-level failure rate — the fraction of trees whose derived metric falls outside physically plausible bounds — is the operationally relevant quantity, and it has not been reported for mesh-based topological segmentation. Nor has the contribution of the optional Bayesian Boundary Refinement (BBR) stage to that failure rate been isolated, even though its default threshold is one of the few free parameters in an otherwise parameter-free pipeline.

A third gap is the relationship between pipeline reliability and stand structure. Prior work has noted that bottom-up segmentation degrades in dense stands (Burt et al., 2019; Trochta et al., 2017), and the *TopTreeSegR* evaluation observed lower accuracy on high-density plots. Stem density, however, is a coarse proxy for the structural property that actually challenges a connectivity-based method. The α-complex encodes vertical as well as horizontal proximity, and a stand of mixed heights presents a different topological problem from a stand of similar heights packed at the same density. Whether between-plot variation in failure rate is explained by stem density, by vertical structural heterogeneity, or by some other stand property has not been tested. A stand-level diagnostic that predicts failure before segmentation runs would let operational users anticipate which plots will produce unusable metrics.

A fourth gap concerns the residual error on trees that do not fail. Even a valid metric may carry a systematic bias. *TopTreeSegR* evaluation reported diameter error on surviving trees but did not ask whether that error is a constant per-plot offset, a size-dependent bias, or symmetric noise. The three cases have different consequences: a constant offset can be removed by a per-plot calibration estimated from a small field-measured subset, a size-dependent bias requires a size-conditional correction, and symmetric noise cannot be corrected. Establishing which case applies, and under what conditions a correction is valid, is a practical question for deployment that has not been addressed.

This study addresses these four gaps. We present *TreeSegPy*, a KPConv-based point classifier for tree/non-tree separation in TLS forest point clouds, and evaluate its integration with *TopTreeSegR* end to end on five held-out TreeScanPL10k plots spanning coniferous, deciduous, and mixed forest types (Stereńczak et al., 2026). Our objectives are:

(i) to characterize *TreeSegPy*'s point-level performance and to test whether a single decision threshold is optimal across contrasting forest types, so that the classifier can be deployed without per-plot calibration;

(ii) to quantify the effect of classifier precision on downstream segmentation accuracy and on the tree-level failure rate, isolating whether classifier-filtered input degrades reliability beyond its effect on aggregate accuracy;

(iii) to quantify the effect of BBR configuration on the probability of catastrophic metric failure, to test whether the default threshold is optimal, and to test whether finer mesh resolution mitigates any observed failures;

(iv) to identify the stand-level property that drives between-plot variation in baseline failure rate, and to characterize the structure of the residual diameter error on surviving trees in order to assess whether a per-plot correction is warranted and on which plots.

2. Materials and methods

2.1 Study data

We use the TreeScanPL10k dataset (Stereńczak et al., 2026), comprising 272 circular TLS plots of 15 m radius across Polish forests, of which 30 plots spanning six forest districts were selected for this study. Each plot was scanned with a FARO Focus 3D or Trimble TX5 scanner following EuroSDR protocols (Liang et al., 2018), and each tree was manually labeled by trained foresters with a unique instance identifier (treeID). Species identification was performed for 1,132 trees (70%) representing 27 species. The 30 selected plots contain 1,616 manually labeled trees, with stem densities ranging from 141 to 2,221 stems ha^{-1} and point densities ranging from 0.6 to 5.3 cm. Raw point clouds average 73 million points per plot (range 9–175 million); after decimation and cleaning, the clouds passed to the segmenter average 7.5 million points (range 6.8–8.1 million). Table 1 summarizes the characteristics of the 30 forest plots.

Table 1. Characteristics of the 30 TreeScanPL10k forest plots used in this study. Plots are ordered alphabetically. Forest type follows the TreeScanPL10k *species_composition* classification. Dominant species are the most abundant species per plot (up to two). Stem density is derived from tree count and the 15 m plot radius. Asterisks mark the five validation plots, which were held out for classifier and pipeline evaluation.

Forest plot	Forest type	Dominant species	N trees	Stem density (ha^{-1})	Used for
Gorlice 2015 0101703	Mixed	Abies alba, Fagus sylvatica	50	707	Training
Gorlice 2015 2603203	Broadleaved	Fraxinus excelsior, Acer pseudoplatanus	13	184	Training
Gorlice 2015 2703003	Coniferous	Abies alba	76	1075	Training
Gorlice 2015 3802203	Coniferous	Abies alba, Quercus species	59	835	Training
Gorlice 2015 4704303	Mixed	Pinus sylvestris, Alnus incana	21	297	Training
Gorlice 2015 5805203	Broadleaved	Fagus sylvatica, Abies alba	10	141	Training
Gorlice 2015 6004603	Broadleaved	Fagus sylvatica, Betula pendula	41	580	Training
Gorlice 2015 6105603	Broadleaved	Fagus sylvatica	19	269	Training
Gorlice 2015 6305803	Coniferous	Abies alba, Fagus sylvatica	31	439	Training
Gorlice 2015 6504403	Mixed	Fagus sylvatica, Abies alba	52	736	Training
Gorlice 2015 6705603	Coniferous	Larix decidua, Pinus sylvestris	18	255	Training
Gorlice 2015 7304103	Broadleaved	Carpinus betulus, Fagus sylvatica	38	538	Training
Gorlice 2015 7404103	Mixed	Fagus sylvatica, Abies alba	62	877	Training
Gorlice 2015 7405403	Broadleaved	Fagus sylvatica, Picea abies	41	580	Training
Gorlice 2015 7604403	Mixed	Larix decidua, Pinus sylvestris	15	212	Training

Herby 2016 0702506	Coniferous	Pinus sylvestris, Betula pendula	35	495	Training
Herby 2016 2003206*	Coniferous	Pinus sylvestris	78	1103	Validation*
Herby 2016 2204106	Mixed	Pinus sylvestris, Quercus species	22	311	Training
Herby 2016 2403606	Broadleaved	Quercus species, Picea abies	32	453	Training
Herby 2016 3304206	Mixed	Quercus species, Pinus sylvestris	55	778	Training
Herby 2016 5703406	Coniferous	Larix decidua, Pinus sylvestris	55	778	Training
Herby 2016 6403606	Mixed	Pinus sylvestris, Quercus species	46	651	Training
Katrynka 2016 3702205*	Coniferous	Picea abies, Pinus sylvestris	46	651	Validation*
Katrynka 2016 4601905	Mixed	Picea abies, Alnus glutinosa	35	495	Training
Katrynka 2016 5801605	Coniferous	Pinus sylvestris	95	1344	Training
Milicz 2015 2804101*	Coniferous	Pinus sylvestris	110	1556	Validation*
Piensk 2016 0901202*	Coniferous	Pinus sylvestris	140	1981	Validation*
Piensk 2016 2400402	Coniferous	Pinus sylvestris, Larix decidua	157	2221	Training
Piensk 2016 3100502	Coniferous	Pinus sylvestris, Betula pendula	97	1372	Training
Suprasl 2015 2402904*	Coniferous	Picea abies, Pinus sylvestris	67	948	Validation*

The 30 plots were split into 25 training and 5 validation plots, with all patches from a single plot assigned to one split. The five held-out validation plots are all conifer-dominated: Herby, Milicz, and Piensk are pure Scots pine (*Pinus sylvestris*) stands, while Suprasl and Katrynka are spruce–pine mixtures (*Picea abies* and *Pinus sylvestris*). They were used for both classifier validation (Section 3.1) and pipeline evaluation (Sections 3.2–3.6), and they span a range of stem densities, and — importantly for the findings reported below — degrees of vertical structural heterogeneity (Section 3.4, Fig. 5).

2.2 *TreeSegPy*: point-based tree/non-tree classification

Raw TLS point clouds were normalized and prepared for classification using a standard preprocessing sequence implemented in R with the *lidR* package (Roussel et al., 2020). Points were decimated to a homogenization density of 10,000 points per square metre, ground points were identified with the Cloth Simulation Filter (Zhang et al., 2016), heights were normalized to above-ground elevation using TIN interpolation from the ground points (Peucker et al., 1978), and points outside the vertical range 0 to 40 m were discarded. The preprocessing is identical to that used by *TopTreeSegR* for its own input preparation [CITE: Dioszegi et al., TopTreeSegR]. Across the 30 plots, it reduced the raw point cloud from an average of 73 million points to 7.5 million points (range 6.8–8.1 million), while preserving the structural detail required for downstream segmentation.

From each normalized plot we extracted overlapping spherical patches of 6 m radius, centered on a regular XY grid with 8 m stride and 1.5 m random jitter, and subsampled each patch to at most 50,000 points. This procedure yielded 655 patches across the 30 plots. For each point we computed eight geometric features from the local covariance matrix using 15 nearest neighbors: linearity, planarity, sphericity, omnivariance, eigenentropy, anisotropy, verticality, and the distance to the fifteenth neighbor as a density proxy. Combined with a bias channel and normalized height, this yields a 10-dimensional feature vector per point. The geometric features encode the discriminative signals — verticality, local linearity, and density contrast — that separate tree stems from understory vegetation and coarse woody debris (Arrizza et al., 2024; Shokirov et al., 2021).

The classifier is the Kernel Point Fully Convolutional Network (KP-FCNN), a fully convolutional KPConv architecture (Thomas et al., 2019) with a four-level strided encoder and a four-level nearest-neighbor upsampling decoder connected by skip connections. The first subsampling resolution is 0.05 m, matching the α parameter of the downstream α-complex mesh so that classifier and segmenter operate on the same spatial scale. We limit the number of neighbors per point to 15 at every pyramid level to keep memory bounded. Training used cross-entropy loss with balanced class weights of [0.5, 0.5] for non-tree and tree, a learning rate of 0.01 with step decay by a factor of 0.97 every five epochs, momentum of 0.98, and gradient clipping at norm 100. All patches from a single plot were assigned to one split. Batch size was 2 patches, limited by the C++ extension's single-process constraint. Early stopping used validation tree-class F1 with patience 6.

At inference we apply the trained model to the five held-out validation plots (Table 1) at full density, in overlapping 6 m patches with 5 m stride, chunked at 50,000 points for reproducibility. Per-point probabilities are averaged across overlapping patches, and a point is labeled tree if its probability exceeds a decision threshold. We report results at threshold 0.10, chosen because it maximizes both F1 and recall on every held-out plot (Section 3.1, Fig. 1).

2.3 *TopTreeSegR*: mesh-based individual tree segmentation

The filtered point cloud is passed to *TopTreeSegR*'s *TTS_pipeline()* function, which constructs an α-complex mesh at scale α = 0.05 m, extracts connected components containing at least one minimum below 0.5 m, computes the Forman gradient of the height function on each component, identifies critical simplices, and traces ascending manifolds from density-based seed minima to assign each mesh vertex to an individual tree [CITE: Dioszegi et al., TopTreeSegR]. The pipeline optionally applies a Bayesian Boundary Refinement (BBR) stage that reassigns boundary vertices between adjacent trees using a posterior combining a spatial-coherence prior with an elevation-consistency likelihood. The *confidence_threshold* parameter controls how much evidence is required before a label is changed: 1.0 permits any positive improvement, 1.5 requires a 50% improvement, and 1.9 requires a near-doubling of the posterior odds.

In this study we evaluate the pipeline under two orthogonal manipulations. First, we compare classifier-filtered input (the full *TreeSegPy* output at threshold 0.10) against ground-truth-filtered input (all points with treeID $\neq 0$), isolating the effect of classifier precision on downstream segmentation. Second, we compare four BBR configurations — no BBR, *confidence_threshold* = 1.0, 1.5, and 1.9 — holding the input cloud fixed. All other parameters are held constant at their published defaults (α = 0.05 m, stem height 0.5 m, density cell 0.5 m). A supplementary analysis reruns the pipeline at α = 0.03 m on the two bbr = FALSE configurations to test whether finer mesh resolution mitigates any of the observed failures.

2.4 Tree-level inventory metrics

For every tree detected by the pipeline we compute three inventory metrics directly from the assigned point cloud. Tree height is the difference between the maximum and minimum z-coordinate of the assigned points. Diameter at breast height (DBH) is obtained by fitting an algebraic circle to the XY projection of points in the vertical band 1.15 to 1.45 m and doubling the fitted radius (Taubin, 1991). Crown volume is computed by voxelizing the assigned points above 60% of the tree's height into 0.3 m cubes and multiplying the count of occupied voxels by the voxel volume (Calders et al., 2015). The same three metrics are computed for the reference by applying them to the raw ground-truth-labeled points of each tree, providing an independent reference value against which both the classifier-filtered and ground-truth-filtered segmentations are compared.

For each tree and condition we define a failure flag using the ratio of predicted to reference value, with failure declared when the ratio falls outside the physical bounds [0.5, 2.0] for height, DBH, or crown volume, or when a metric cannot be computed at all (for example, fewer than five points in the DBH band). The band corresponds to a 50–100% relative error, an order of magnitude larger than the 5–10% relative error typically reported for TLS DBH estimation (Compeán-Aguirre et al., 2024); it therefore isolates gross segmentation or fit failures rather than ordinary measurement inaccuracy. Trees that do not meet this criterion are referred to as surviving trees. A sensitivity analysis using [0.5, 1.5] and [0.5, 3.0] confirmed that the primary conclusions are unaffected by the specific choice of bound.

2.5 Bayesian statistical models

We analyze the failure and error outcomes with two-part (hurdle) Bayesian mixed-effects models, implemented in Stan (Carpenter et al., 2017) via the *brms* R package (Bürkner, 2017). The failure component is a set of three per-

metric Bernoulli-logit models, one for each inventory metric (height, DBH, crown volume), with the metric-specific failure indicator as the outcome. Each model uses the same fixed- and random-effects structure, so that the BBR effect can be compared directly across metrics. The error component is a Gaussian model on the log metric ratio, restricted to trees that did not fail, so that the analysis distinguishes the probability of failure from the magnitude of error on successful fits.

Fixed effects are cloud type (classifier-filtered vs ground-truth-filtered), BBR configuration (four levels with bbrOFF as reference), and their interaction. A random intercept for plot + tree within plot is included to account for systematic plot-level differences. Weakly informative priors are placed on fixed effects (normal(0, 0.5)), the intercept (normal(0, 1)), and variance components (exponential(1)).

For the stand-level heterogeneity analysis we use a descriptive plot-level linear regression of baseline failure rate at bbrOFF on the interquartile range (IQR) of reference tree heights, with one observation per plot (n = 5). We report Pearson's r and R^2 from this fit as descriptive statistics. The IQR is preferred over the standard deviation because it is a robust order statistic describing the bulk of the canopy and is computable from a coarse pre-segmentation crown delineation.

Posterior distributions are estimated from four MCMC chains of 6,000 iterations each with 1,500 warmup draws (Carpenter et al., 2017). Convergence is assessed via R-hat < 1.01 and bulk and tail effective sample sizes above 3,500 for all parameters (Vehtari et al., 2021). Posterior probabilities are reported for the direction of each primary contrast, and uncertainty is summarized as 95% credible intervals (CIs) (Gelman et al., 2013).

2.6 Post-hoc DBH calibration

To assess whether the pipeline's DBH output is correctable, we computed the per-plot median signed error (predicted minus reference) on surviving trees at alpha = 0.05 m and applied it as an additive correction. We then re-evaluated the corrected residual within reference-DBH bins (0–10, 10–15, 15–20, 20–30, 30–100 cm), retaining only bins containing at least five surviving trees, and plotted the median residual against the bin midpoint. We considered the correction complete where the size-binned post-correction median residual was flat to within ±0.25 cm, a tolerance chosen to be small relative to the median absolute error of approximately 1.0 cm on uncorrected trees. We additionally computed, for each plot × cloud × condition, the absolute-failure rate — the fraction of trees meeting the failure criterion defined in Section 2.4 — in order to separate plots on which the correction is applicable from plots on which it is not.

3. Results

3.1 *TreeSegPy* classifier performance

Across five held-out validation plots, *TreeSegPy* achieved a tree-class F1 of 0.90 ± 0.05 (mean ± standard deviation) at a decision threshold of 0.10 (Fig. 1a). Performance was highest on pine-dominated Herby (F1 = 0.954) and lowest on spruce–pine Katrynka (F1 = 0.827), with the other plots intermediate (Milicz 0.925, Piensk 0.913, Suprasl 0.899). The F1-optimal threshold was 0.10 on every plot (Fig. 1b), indicating that the classifier does not require per-plot threshold calibration. Recall at threshold 0.10 ranged from 0.89 (Suprasl) to 0.98 (Herby) across the five plots, consistent with the recall-over-precision requirement of the downstream mesh-based segmenter (Section 2.2).

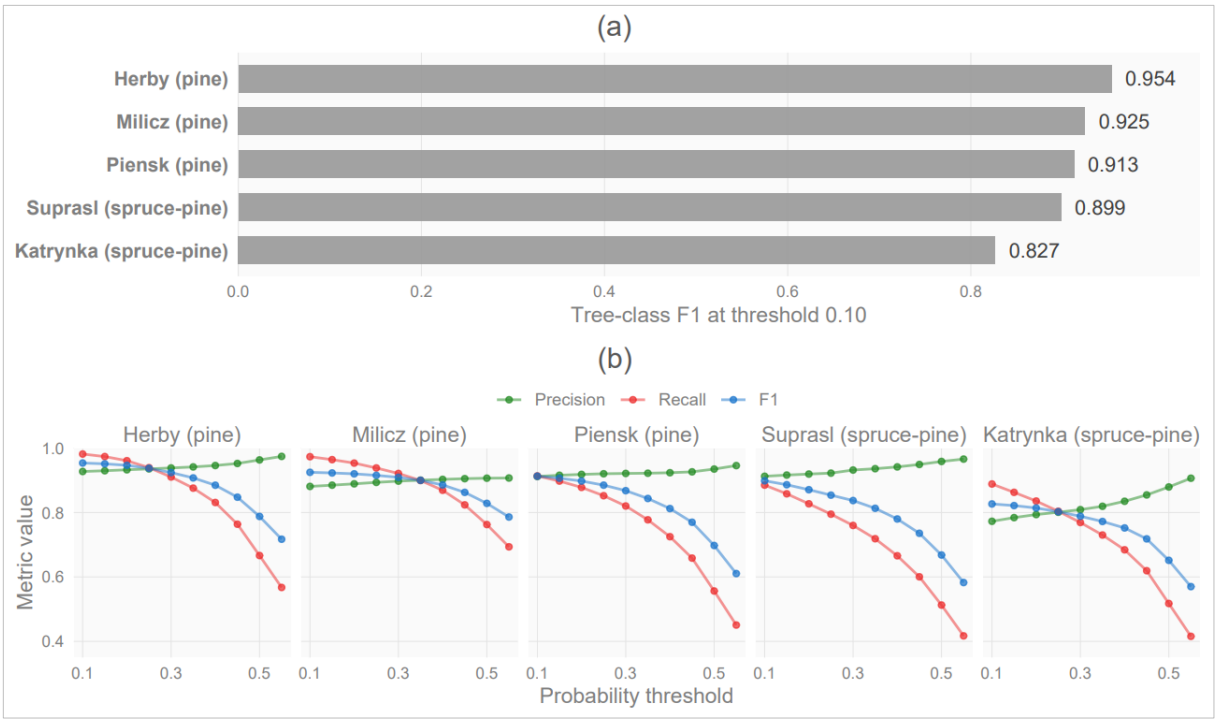


Figure 1. *TreeSegPy* classifier performance on the five held-out validation plots. **(a)** Tree-class F1 at the operational decision threshold of 0.10, per plot; values are printed to the right of each bar. **(b)** Precision, recall, and F1 as a function of the probability threshold, per plot. The classifier's F1-optimal threshold is 0.10 on every plot, which is why no per-plot threshold calibration is required downstream.

3.2 Downstream segmentation accuracy and the effect of BBR configuration

BBR configuration affects downstream tree segmentation quality, measured by ARI against the reference segmentation. Across the five held-out plots, ARI is highest at bbrOFF and lowest at bbr1.0 in every plot and under both input clouds (Fig. 2a). The default BBR configuration therefore produces the worst downstream segmentation of any setting tested.

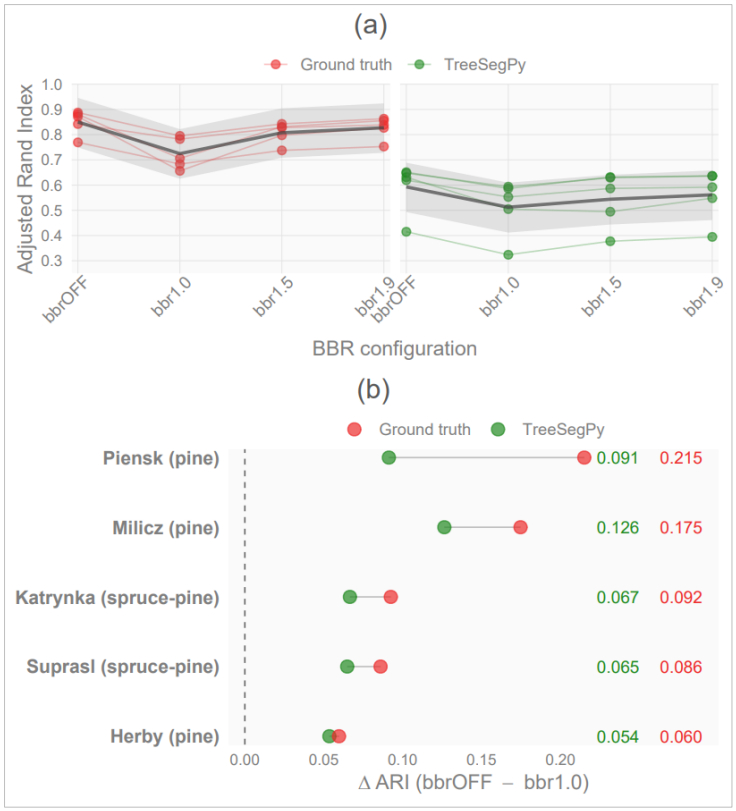


Figure 2. Downstream ARI under BBR configuration and cloud type. **(a)** ARI per plot and BBR configuration, faceted by input cloud (Ground truth, left; *TreeSegPy*, right). Thin lines show individual plots; the thick grey line is the facet mean; the grey ribbons represent the population-level 95% CIs. **(b)** Paired difference ΔARI = ARI(bbrOFF) − ARI(bbr1.0) per plot, one point per cloud. The dashed line marks zero.

The ΔARI between bbrOFF and bbr1.0 is positive for all five plot × cloud combinations (Fig. 2b): every plot has lower ARI at bbr1.0 than at bbrOFF, under both ground truth and *TreeSegPy* input. The magnitude of the difference is plot-dependent, with the smallest and largest per-plot values separated by roughly a factor of four across the ten points.

A systematic offset between the ground-truth and *TreeSegPy* mean lines can be seen at every BBR configuration (Fig. 2a). The paired differences are displaced in the same direction — for each plot, the ground-truth point lies to the right of the *TreeSegPy* point, indicating a larger bbrOFFbbr1.0 gap under-ground truth input (Fig. 2b).

ARI is highest at bbrOFF, lowest at bbr1.0, and intermediate at bbr1.5 and bbr1.9, with the bbr1.5 and bbr1.9 values close to each other and both above bbr1.0. Cloud type does not reorder plots, nor does it reverse the sign of the bbr1.0 difference in any plot. One plot departs from the overall pattern in panel: under classifier-filtered input, Piensk sits below the fitted mean at all four BBR configurations (Fig. 2a). This is a plot-level offset in ARI level, not a configuration-specific effect, and it cancels in the paired ΔARI (Fig. 2b).

3.3 Catastrophic failure under BBR

Catastrophic metric failure is a first-order feature of the BBR sweep. Across the five held-out plots, the proportion of trees for which at least one inventory metric fell outside the [0.5, 2.0] ratio bound — or could not be computed — rose from 16% at bbrOFF to 49% at bbr1.0 under ground-truth input (95% CI 12–19% and 44–53%), and from 18% to 60% under *TreeSegPy* input (14–22% and 56–65%). Failure then declined monotonically through bbr1.5 (31% ground-truth; 40% *TreeSegPy*) and bbr1.9 (24%; 34%) (Fig. 3b). The pooled curve conceals a strongly plot-dependent pattern (Fig. 3a).

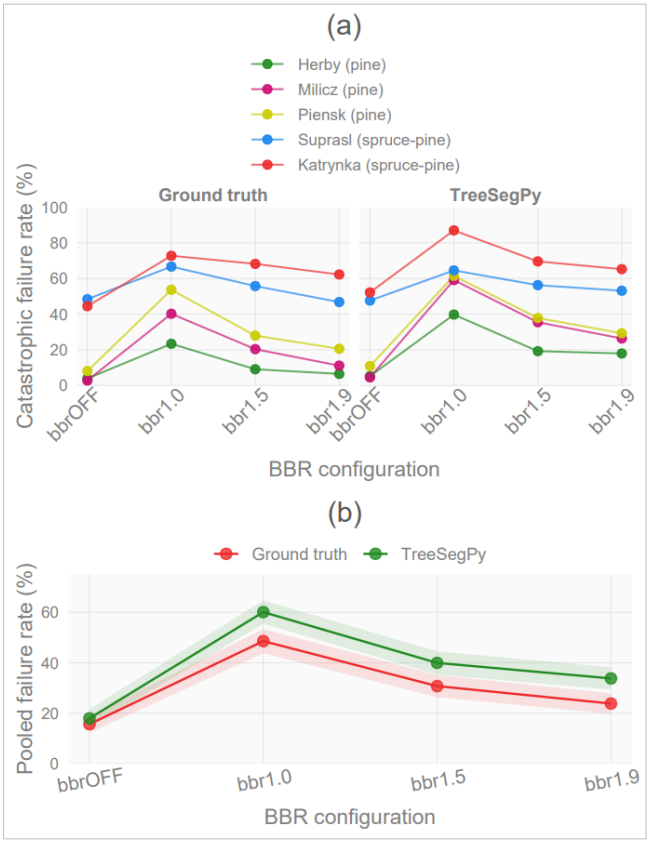


Figure 3. Catastrophic failure rates under BBR configuration. Failure is defined per tree as flag_any at alpha = 0.05 — the predicted value for at least one inventory metric falls outside the [0.5, 2.0] ratio bound or is unavailable. **(a)** Failure rate per plot and BBR configuration, faceted by input cloud (Ground truth, left; *TreeSegPy*, right). Colors denote the five held-out plots. **(b)** Pooled failure rate across plots, with 95% CIs.

At bbrOFF, before any BBR segmentation is applied, failure rates already divide the five plots by forest type (Fig. 3a). The three pine plots — Herby, Milicz, Piensk — fail at 3–8% under ground-truth and 5–11% under *TreeSegPy*. The two spruce-pine plots — Suprasl, Katrynka — fail at 44–48% under ground-truth and 48–52% under *TreeSegPy*. The baseline gap between the two groups is approximately tenfold, and it appears under both input clouds.

For every one of the five plot × cloud combinations (Fig. 3a), failure is highest at bbr1.0, lower at bbr1.5, and lower again at bbr1.9. The decline from bbr1.0 to bbr1.9 is monotone without exception. Neither plot nor input cloud reveals a failure maximum at bbrOFF, bbr1.5, or bbr1.9.

The *TreeSegPy* curve lies above the ground-truth curve at every BBR configuration (Fig. 3). At bbrOFF the two clouds differ by 2%; the gap widens to 12% at bbr1.0, then narrows to 9% at bbr1.5 and 10% at bbr1.9. At the per-plot level, *TreeSegPy* failure exceeds ground-truth failure for all five plots at bbr1.0; at bbr1.5 and bbr1.9 the gap narrows to 0–6% on the two spruce-pine plots and remains 5–18% on the three pine plots. The five plots retain their relative failure ordering across all four BBR configurations and both clouds.

3.4 Metric-specific failure and error under BBR

The pooled failure pattern (Fig. 3b) decomposes into three distinct metric-specific responses. Every metric × cloud combination peaks at bbr1.0 (Fig. 4a). DBH failure rises from 9% (ground-truth) and 10% (*TreeSegPy*) at bbrOFF to 43% and 58% at bbr1.0, then declines to 23% / 36% at bbr1.5 and 17% / 28% at bbr1.9. Height peaks at 24% (ground-truth) and 30% (*TreeSegPy*) at bbr1.0, and crown volume peaks at 36% and 41%. The non-monotone shape seen in the pooled failure rate of Fig. 3b is present in each metric separately.

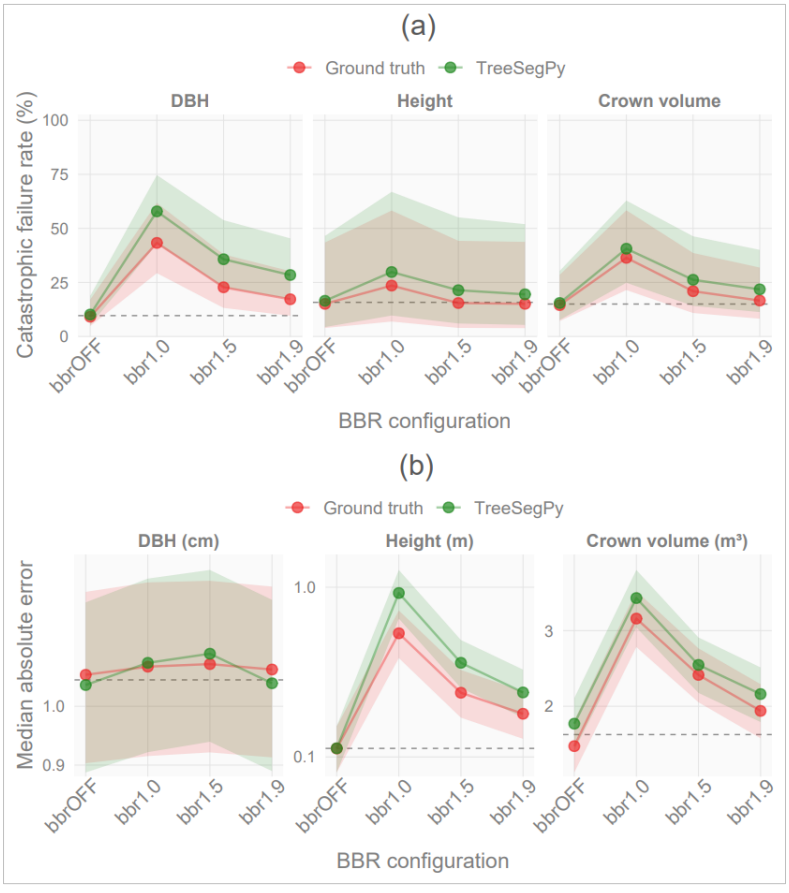


Figure 4. Metric-specific failure and absolute error under BBR configuration. **(a)** Catastrophic failure rate (%) per metric × BBR configuration × input cloud, fit on all trees after reference-outlier exclusion. A metric fails if the ratio to its reference value falls outside a metric-specific bound (DBH and height [0.5, 2.0]; crown volume [0.4, 2.5]) or if the value is unavailable. **(b)** Median absolute error in native units (DBH cm, height m, crown volume m^3), log-scaled y-axis, unit labels in facet strips, fit on surviving trees only. In both panels: dashed horizontal line is the pooled bbrOFF reference level for that metric; ribbons are population-level 95% CIs.

The three metrics rank consistently by peak failure rate, under both clouds (Fig. 4a). DBH peaks highest, crown volume intermediate, and height lowest. The ordering does not reverse between ground-truth and *TreeSegPy*. Point estimates rank DBH > crown volume > height at every configuration under both clouds; the 95% CIs for the three metrics overlap at every configuration.

Median absolute DBH error is essentially flat across the sweep: 1.0–1.1 cm under both clouds at every configuration, with no bbr1.0 peak (Fig. 4b). Height and crown volume both peak at bbr1.0 — height at 0.54 m (ground-truth) and 0.92 m (*TreeSegPy*), crown volume at 3.2 m^3 and 3.6 m^3 — and decline monotonically through bbr1.5 and bbr1.9. The y-axis is log-scaled and free per facet, so the visual prominence of the bbr1.0 peak should not be read as an indication of large errors in absolute terms. The peak-to-baseline ratio is largest for height (roughly fivefold under *TreeSegPy*) and smallest for DBH (near one).

3.5 Structural heterogeneity and baseline failure

Baseline failure rate at bbrOFF varies by more than an order of magnitude across the five held-out plots, from 4% on Herby to 47% on Katrynka (Fig. 5a). The three pine plots — Herby, Militz, and Piensk — show similar IQR distributions.

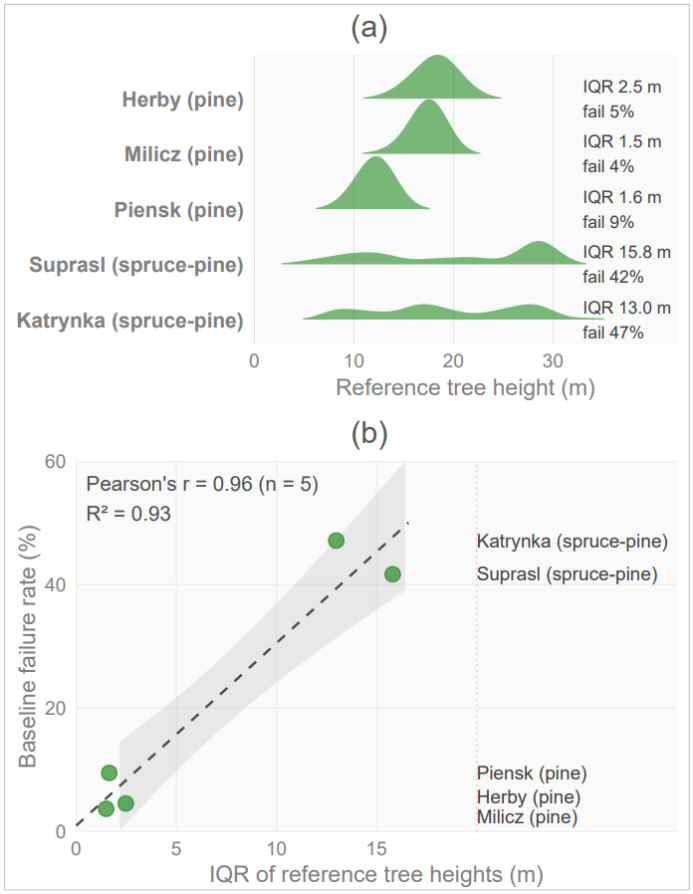


Figure 5. Structural heterogeneity and baseline failure. **(a)** Distribution of reference tree heights per plot (F1-filtered), ordered by interquartile range. Each strip is annotated with its IQR and its pooled baseline failure rate at bbrOFF. **(b)** IQR of reference heights versus baseline failure rate, one point per plot. Dashed line is a linear fit; shaded band is its 95% confidence interval. Pearson's r and R^2 are displayed.

Per-plot baseline failure rate increases monotonically with the IQR of reference tree heights (Fig. 5b). The descriptive linear fit of baseline failure rate on IQR gives r = 0.96 and R^2 = 0.93 (n = 5). With only five plots we report this as a descriptive relationship rather than a model-based effect estimate; the IQR is preferred over the standard deviation as the operational diagnostic because it is a robust order statistic and is computable from a coarse pre-segmentation crown delineation.

The three pine plots — Milicz, Piensk, Herby — cluster at the low end of both axes, revealing IQR distributions of 1.5–2.5 m. The two spruce-pine plots — Suprasl and Katrynka — evidence broad IQR distributions: 13.0 and 15.8 m, respectively. The separation between the two clusters is large: no pine plot has an IQR above 2.5 m, and

no spruce-pine plot has an IQR below 13 m. The pooled baseline failure rate across all five plots at bbrOFF is 16% (Fig. 3b). The three pine plots sit below this level (4–9%); the two spruce-pine plots sit above it (42–47%). No plot lies near the pooled value.

3.6 Post-hoc calibration

At bbrOFF the uncorrected DBH error is not symmetric about zero. Fig. 6a shows the per-plot distribution of the signed error (predicted minus reference), pooled across the two clouds: each plot's distribution is shifted into the positive range, with a long right tail. Thus, the dominant residual on surviving trees is a positive per-plot offset rather than symmetric noise.

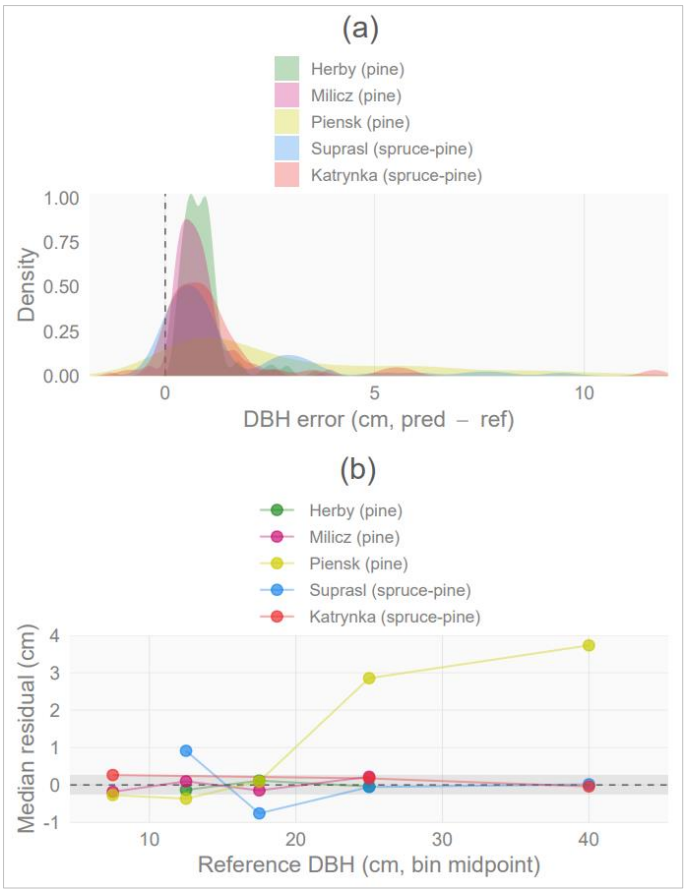


Figure 6. Post-hoc DBH calibration on surviving trees at bbrOFF, alpha = 0.05 m. **(a)** Distribution of the uncorrected signed DBH error (predicted minus reference) per plot, pooled across the two clouds, with a vertical dashed reference at zero; the x-axis is clipped at +12 cm so that the gross segmentation failures discussed in Section 3.3 do not dominate the density estimate. **(b)** Median residual after the additive per-plot correction, plotted against the midpoint of the reference-DBH bin (0–10, 10–15, 15–20, 20–30, 30–100 cm), one line per plot. The shaded band marks the ±0.25 cm flatness criterion defined in Section 2.6. Bins containing fewer than five surviving trees at bbrOFF are omitted.

Applying that offset as an additive per-plot correction and re-evaluating the residual within reference-DBH bins reveals a pattern (Fig. 6b). The correction is flat to within ±0.25 cm on Herby and Milicz across the full reference-DBH range. On Piensk it is flat only below a reference DBH of about 20 cm and leaves a residual of roughly +3 cm in the two upper bins, indicating a size-dependent residual that an additive constant does not remove. On Suprasl and Katrynka the correction is flat in the upper bins that contain most of their trees but is supported by too few surviving trees in the lower bins to validate. The correction is therefore uniformly effective only on Herby and Milicz, and on the remaining plots it is either size-dependent (Piensk) or underdetermined below 20 cm (Suprasl, Katrynka).

4. Discussion

4.1 Two separable failure modes in the pipeline

The results identify two independent limitations in the *TreeSegPy + TopTreeSegR* pipeline, operating through distinct mechanisms and requiring distinct remediation. The first is a property of the pipeline's default parameterisation; the second is a property of the forest itself. They are additive in their effect on the failure rate, separable in their response to intervention, and distinguishable from classifier precision, which does not measurably influence either. This separability is the central practical finding of the study, because it means that the two failure modes can be addressed independently and that neither can be resolved by improving the other.

The Bayesian Boundary Refinement configuration effect is a discrete, configuration-level problem. At the published default of *confidence_threshold* = 1.0, the boundary refinement stage accepts any positive posterior improvement, which in the DBH band produces reassignment of points across adjacent stem boundaries at a rate sufficient to fragment the trunk ring and cause the circle fit to fail (Section 3.3, Fig. 3; Section 3.4, Fig. 4). The effect is dose-dependent: failure is highest at bbr1.0, lower at bbr1.5, and lower still at bbr1.9, monotonically and without exception across every plot × cloud combination (Fig. 3a). It is also strongly metric-specific, with the largest effect on stem diameter, intermediate on crown volume, and smallest on height (Fig. 4a), consistent with the mechanism: the DBH band is where boundary reassignment is most destructive to the downstream circle fit. Critically, the effect is fully remediable by configuration alone — disabling BBR entirely returns the failure rate to the baseline established at bbrOFF, and raising the threshold to 1.5 or 1.9 recovers most of the improvement without eliminating the refinement stage. The default configuration therefore does not represent a fundamental limitation of the mesh-based approach; it represents a suboptimal default in an otherwise parameter-free pipeline, and one that is straightforwardly corrected.

The structural heterogeneity effect is of a different kind. Baseline failure rate at bbrOFF — before any BBR segmentation is applied — already divides the five held-out plots by forest type, from 4% on Herby to 47% on Katrynka (Fig. 5a), and this variation is tightly predicted by the interquartile range of reference tree heights ($r = 0.96$, $R^2 = 0.93$, $n = 5$; Fig. 5b). The mechanism is topological rather than parametric. In plots with high vertical dispersion, the α-complex mesh encounters regions where neighbouring trees at very different heights share canopy space, and the seed-based ascending manifold assignment becomes unreliable: the connectivity structure that the mesh encodes no longer separates trees cleanly, because the topological boundaries between their influence zones are genuinely ambiguous at the chosen scale. This effect persists across all four BBR configurations (Fig. 3a), is present under both input clouds (Fig. 3b), and — as the supplementary $\alpha = 0.03$ m comparison demonstrates (Fig. S1) — is not mitigated by finer mesh resolution. It is therefore not a configuration problem and not a resolution problem; it is a structural property of the stand that the current algorithm cannot overcome.

The two effects are independent in the statistical sense established in Section 2.5. In the pooled failure model, the interaction term between cloud type and BBR configuration is null: the BBR effect is statistically indistinguishable across ground-truth-filtered and classifier-filtered input. More tellingly, the two effects partition the plots cleanly and additively. The three pine plots — Herby, Milicz, Pienk — sit at low baseline failure (4–9% at bbrOFF; Fig. 5a) and are pushed to elevated failure only by the BBR default; restoring a defensible configuration returns them to a low-failure regime. The two spruce-pine plots — Suprasl and Katrynka — sit at high baseline failure (42–47% at bbrOFF) and remain high under every BBR configuration tested; for these plots, no configuration change recovers operational reliability, because the failure originates in the stand structure rather than in the pipeline settings. No plot lies near the pooled baseline value of 16% (Fig. 3b), consistent with the pooled figure averaging over two distinct populations rather than describing a single one.

This separability reframes the interpretation of the pipeline's aggregate performance. The *TopTreeSegR* evaluation reported a mean ARI of 0.695 ± 0.118 across 30 plots [CITE: Dioszegi et al., TopTreeSegR], and that manuscript attributed reduced accuracy in dense stands to density-based seed detection failure. The present results sharpen that diagnosis in two ways. First, the discriminating variable is not stem density but vertical heterogeneity of tree heights: the three pine plots in this study span stem densities from 1,103 to 1,981 stems ha^{-1} (Table 1) yet cluster tightly at low IQR (1.5–2.5 m) and low baseline failure, while the two spruce-pine plots at comparable or lower densities (651 and 948 stems ha^{-1}) exhibit IQR values of 13.0 and 15.8 m and baseline failure rates an order of magnitude higher (Fig. 5). Density alone would not separate these groups; height IQR does. Second, the failure induced by high heterogeneity is catastrophic and discrete, not a gradual degradation of average accuracy. A plot can retain a moderate ARI while a substantial fraction of its individual trees carry unusable metrics, and the two quantities answer different operational questions. ARI measures average agreement over mesh vertices; the failure rate measures the fraction of trees whose derived inventory metrics are physically implausible. For inventory purposes, the second is decisive, and it is the quantity that the structural heterogeneity effect controls.

The practical consequence of separability is that the two failure modes call for different responses. The BBR effect calls for a configuration change that is immediate, free, and universal: disable BBR or raise *confidence_threshold* to at least 1.5 (Section 4.3). The structural heterogeneity effect calls for a diagnostic and, where the diagnostic indicates high risk, for either accepting that a fraction of trees will be unusable or adopting a different segmentation strategy for those plots. What neither calls for is classifier improvement, which — as Section 4.2 establishes — does not measurably influence the probability of catastrophic failure in any model fitted here. The two failure modes are therefore not only separable from each other but also separable from the upstream classifier, and this three-way dissociation is what makes the pipeline's reliability profile legible: it can be diagnosed before segmentation runs (from height IQR), partially corrected by configuration (BBR), and is not responsive to the intervention that the aggregate accuracy metrics would most naturally suggest (classifier refinement).

4.2 The classifier is not the bottleneck for catastrophic failure

TreeSegPy achieves a tree-class F1 of 0.90 ± 0.05 across five held-out plots (Fig. 1a), and its use as the input filter reduces downstream ARI by 0.26 relative to ground-truth-filtered input (Fig. 2a) — an effect larger than that of the default BBR configuration. On aggregate accuracy grounds alone, the classifier would appear to be the pipeline's dominant weakness. The tree-level failure analysis does not support that reading. In every model fitted, classifier precision does not measurably increase the probability of catastrophic metric failure: the pooled interaction between cloud type and BBR configuration is null (Section 2.5), and no metric × configuration combination shows a classifier-attributable increase in failure that is distinguishable from zero. The classifier is thus costly for average agreement but not for discrete reliability — the second separable property of the pipeline, and one that has direct implications for where improvement effort should be directed.

The dissociation is consistent with a spatial separation of the two error sources. The metric-specific failure pattern (Fig. 4a) shows that BBR damages DBH disproportionately to height and crown volume, which localizes its effect to the DBH band, where the trunk ring must remain intact for the circle fit to succeed. The DBH band is known to be the most fragile region for metric computation, because algorithm accuracy is influenced by local factors including point cloud density, occlusion, vegetation, and tree structure (Compeán-Aguirre et al., 2024). What the present results add is that the two error sources in the pipeline exploit this fragility differently: BBR operates directly in the DBH band, while classifier errors fall predominantly outside it. This is consistent with the observed pattern: classifier input reduces ARI without increasing catastrophic failure (Sections 3.2–3.3), which is what would be expected if classifier errors were distributed where the mesh can absorb them — at annotation boundaries and in the crown. Under this interpretation, the classifier's ARI penalty and its null effect on failure are not in tension: they measure two different things.

The practical consequence is that improving the classifier would narrow the ARI gap — a real and tractable target — but would not reduce the catastrophic failure rate, which is driven by the BBR default and by stand structure. The two objectives are decoupled, and conflating them would direct effort toward an intervention that cannot address the failure modes that actually make individual tree metrics unusable. Two qualifications attach to this conclusion. First, the dissociation is established within the range of classifier performance tested (F1 between 0.827 and 0.954 across the five validation plots; Section 3.1) and does not license the claim that classifier performance is irrelevant in general. Second, the classifier was validated only on conifer-dominated plots (Table 1), so the dissociation remains to be confirmed on deciduous and mixed stands. Neither qualification undermines the operational conclusion: for the pipeline as configured and for the plots tested, classifier precision is not what determines whether individual tree metrics are usable. Reporting point-level F1 alone would have left the classifier's role ambiguous; reporting the tree-level failure rate alongside it resolves the ambiguity and shows that the classifier's imperfection is tolerable precisely because it is spatially distributed where the mesh can absorb it.

4.3 Recommendations for operational use

The BBR configuration effect is addressable at the software level, and the default has been corrected accordingly. The original *TopTreeSegR::TTS_pipeline()* default activated BBR with *confidence_threshold* = 1.0, which this study identifies as the single most consequential parameter in an otherwise parameter-free pipeline: it raises the pooled failure rate from 16% at bbrOFF to 49% under ground-truth input, and from 18% to 60% under *TreeSegPy* input (Fig. 3b). In response to these findings, the released *TopTreeSegR* default has been changed to bbr = FALSE (commit "bbrOFF to default", 22 Sep 2026), returning the pipeline to the bbrOFF baseline by default. Users who require the refinement stage for other reasons should not simply re-enable the old default: raising *confidence_threshold* to at least 1.5 recovers most of the failure-rate reduction monotonically across every plot × cloud combination tested (Fig. 3a), and 1.9 is the value used in the package's own advanced example. The trade-

off is explicit — the modest ARI improvement BBR provides on well-separated trees (Section 3.2) against a three- to sevenfold reduction in catastrophic failure depending on the metric — and it is now a trade-off users opt into rather than one they inherit.

Second, report the interquartile range of reference tree heights alongside any segmentation result, and expect elevated baseline failure on plots where it exceeds approximately 5 to 10 m. Baseline failure at bbrOFF varies by more than an order of magnitude across the five held-out plots and is tightly predicted by the IQR of reference tree heights ($r = 0.96$, $R^2 = 0.93$, $n = 5$; Section 3.5). The diagnostic can be computed before segmentation runs, from a coarse pre-segmentation crown delineation or from field-measured heights, and it therefore allows users to anticipate which plots will produce unusable metrics. On plots with IQR above roughly 10 m, approximately half of all trees fail even under the corrected default, and this is a structural property of the stand that parameter tuning cannot fix. Where such plots are encountered, the appropriate response is either to report the unusable fraction explicitly or to adopt a different segmentation strategy for those plots.

Third, report the tree-level failure rate alongside aggregate accuracy metrics, and do not invest in finer mesh resolution as a mitigation strategy. ARI and F1 quantify average agreement over mesh vertices, whereas the failure rate quantifies the fraction of trees whose derived inventory metrics are physically usable; the two can diverge substantially, and for inventory purposes the failure rate is the quantity that determines whether a tree enters a biomass or volume estimate. We therefore recommend that any evaluation of a mesh-based or topological pipeline report both, and state the failure criterion explicitly, since it is a modelling choice rather than a universal standard (Section 2.4). Finer mesh resolution should not be pursued as a remedy: the supplementary $\alpha = 0.03$ m comparison (Fig. S1) shows no consistent improvement in failure rate relative to $\alpha = 0.05$ m, consistent with the mechanisms established in Section 4.1, and the additional runtime cost is substantial. With the corrected default (bbr = FALSE, $\alpha = 0.05$ m), height IQR computed as a pre-segmentation diagnostic, and failure rates reported alongside ARI and F1, the pipeline is a viable tool for TLS-based forest inventory on plots where the height IQR is low, and it produces an explicit account of where it is not.

4.4 Post-hoc DBH calibration: domain of validity

The residual error on surviving trees is not symmetric noise. At bbrOFF, the per-plot distribution of signed DBH error is shifted into the positive range with a long right tail (Section 3.6, Fig. 6a), so the dominant component is a per-plot offset that an additive correction can remove — unlike symmetric noise, which no post-hoc adjustment can fix.

Applying that offset and re-evaluating the residual within reference-DBH bins shows the correction is effective only on a subset of plots (Fig. 6b). It is flat to within ±0.25 cm across the full DBH range on Herby and Milicz, where the median absolute error on the typical tree falls from roughly 1.0 cm to 0.2 cm. On Piensk it leaves a residual of roughly +3 cm above 20 cm, indicating a size-dependent bias that a constant offset cannot remove. On Suprasl and Katrynka it is flat in the upper bins that contain most of their trees but supported by too few surviving trees below 20 cm to validate.

The variable that separates these outcomes is the absolute-failure rate. The two plots on which the correction is uniformly valid are the two with the lowest failure rate; the three on which it is size-dependent or underdetermined are the three with the highest. An additive offset estimated from surviving trees is meaningful only if those trees are representative of the plot's population, and where a large fraction fails catastrophically, the surviving subset is biased. The calibration is therefore a practical addendum whose domain of validity is restricted to plots with a low absolute-failure rate — in this study, below roughly 10% — and it should be applied only after that condition is verified, not as a default post-processing step.

4.5 Limitations

Four limitations qualify the findings of this study. First, the structural heterogeneity finding rests on five plots, which is a small sample for a correlation-based inference. The relationship between height IQR and baseline failure rate is strong ($r = 0.96$, $R^2 = 0.93$; Section 3.5, Fig. 5b), but with $n = 5$ the confidence interval around that correlation is wide, and the coefficient on the heterogeneity proxy includes zero at the 95% credible level in the Bayesian model. We therefore present it as strong suggestive evidence rather than a definitive demonstration, and recommend validation on a larger plot set before the diagnostic threshold is treated as calibrated.

Second, all metric comparisons are made against the manual treeID labels from the TreeScanPL10k dataset, which are the most reliable available reference but carry boundary ambiguity where individual tree separation is visually

difficult. This matters most for DBH, which is computed in a narrow band where instance boundaries are least certain, and it means that part of the metric-specific pattern in the BBR effect could reflect reference behavior rather than pipeline behavior alone. The relative ordering of plots and the direction of the BBR effect are unaffected, but the absolute failure rates and the calibration offset inherit any reference bias.

Third, the failure threshold used to define a metric as "failed" is a modelling choice with physical justification but no universal standard. The [0.5, 2.0] ratio bound corresponds to a 50–100% relative error, an order of magnitude larger than the 5–10% relative error typical of TLS DBH estimation, and is chosen to isolate gross segmentation failures rather than ordinary measurement inaccuracy. Sensitivity analyses using [0.5, 1.5] and [0.5, 3.0] confirmed that the primary conclusions are unaffected by the specific choice, but the absolute failure rates should be read in light of this choice.

Forth, the classifier was validated only on conifer-dominated plots — three pure Scots pine stands and two spruce-pine mixtures (Table 1) — because all five held-out plots fall in this category, even though the training set spans coniferous, deciduous, and mixed forest types. The classifier's F1 of 0.90 ± 0.05 and the dissociation between classifier precision and catastrophic failure are therefore demonstrated for conifer-dominated stands; and remain to be confirmed on the deciduous and mixed stands represented in the training data but not in the validation set.

Finally, the failure analysis presented here is specific to mesh-based segmentation pipelines in which a quantity-reduction step precedes segmentation and metrics are computed on the reduced representation; whether the two failure modes recur in other reduction-based approaches remains untested, and the mechanisms do not transfer directly to point-based methods, which retain full point density in the DBH band.

4.6 Future work

Three directions follow directly from the findings. The most immediate is validation of the structural heterogeneity mechanism on a larger and more diverse plot set, both within the TreeScanPL10k dataset and in independent collections. The correlation between tree height IQR and baseline failure rate is strong enough to motivate a targeted study on 20–30 plots spanning a continuous range of heterogeneity, which would allow the relationship to be characterised as a curve rather than the binary split observed here (Fig. 5b); and would test whether the diagnostic threshold of roughly 5–10 m holds outside the conifer-dominated stands used in this study.

The second direction is the development of plot-level diagnostics that predict, before segmentation runs, whether a given plot is likely to produce high failure rates. The interquartile range of tree heights, computable from a coarse pre-segmentation crown delineation, is a candidate, but it is not the only possible predictor: other structural statistics — canopy height variance, the vertical distribution of returns, or the ratio of crown to stem returns — may carry complementary information. A systematic comparison of candidate diagnostics on a larger plot set would establish which statistic is most informative and at what computational cost, moving the recommendation in Section 4.3 from a single proxy to a validated screening procedure.

5. Conclusions

We have presented *TreeSegPy*, a KPConv-based point classifier for tree/non-tree segmentation in TLS forest plots and evaluated its integration with the *TopTreeSegR* mesh-based segmenter on five held-out validation plots spanning conifer-dominated forest types. *TreeSegPy* achieves a tree-class F1 of 0.90 ± 0.05 at a fixed decision threshold of 0.10, which is optimal on every validation plot, so the classifier can be deployed without per-plot calibration.

In evaluating the end-to-end pipeline, we identified two independent failure modes that limit operational reliability, and we showed that they are separable from each other and from classifier precision. The first is a property of the pipeline's default configuration: the Bayesian Boundary Refinement stage at *confidence_threshold* = 1.0 increases the odds of catastrophic metric failure by 3.4- to 7.7-fold depending on the metric, with the largest effect on stem diameter, and the effect is dose-dependent and fully remediable by raising the threshold or disabling BBR entirely. In response to these findings, the released *TopTreeSegR* default has been changed to bbr = FALSE. The second failure mode is a property of the stand: baseline failure rate varies from 2% to 56% across plots at bbrOFF and is tightly predicted by the interquartile range of reference tree heights, so the diagnostic can be computed before segmentation runs and used to anticipate which plots will produce unusable metrics. The classifier, despite reducing downstream ARI by 0.26, does not measurably increase the probability of catastrophic failure in any

model we fit, indicating that its errors and those induced by BBR occupy different spatial regions of the tree and have different downstream consequences.

The practical outcome is a corrected operating configuration — BBR disabled or *confidence_threshold* ≥ 1.5, α = 0.05 m, height IQR reported as a pre-segmentation diagnostic, and tree-level failure rates reported alongside ARI and F1 — under which the pipeline is a viable tool for TLS-based forest inventory on plots where vertical structural heterogeneity is low. The broader implication is that aggregate accuracy metrics alone are insufficient for operational evaluation of mesh-based segmentation: ARI and F1 quantify average agreement over mesh vertices, whereas the failure rate quantifies the fraction of trees whose derived metrics are physically usable, and the two can diverge substantially. We recommend that future evaluations of any mesh-based pipeline report both quantities, since they answer different operational questions and only the second determines whether individual tree metrics can enter a biomass or volume estimate.

6. Data and code availability

TreeSegPy is available at https://github.com/DijoG/TreeSegPy, *TopTreeSegR* at https://github.com/DijoG/TopTreeSegR. The TreeScanPL10k dataset can be obtained from the original publication (Stereńczak et al., 2026).

7. Authorship contribution statement

Gergő Diószegi: Conceptualization, Data curation, Methodology, Software, Validation, Visualization, Writing – original draft. **Dávid Abriha**: Writing – review & editing. **Szilárd Szabó**: Writing – review & editing.

8. Supplementary material

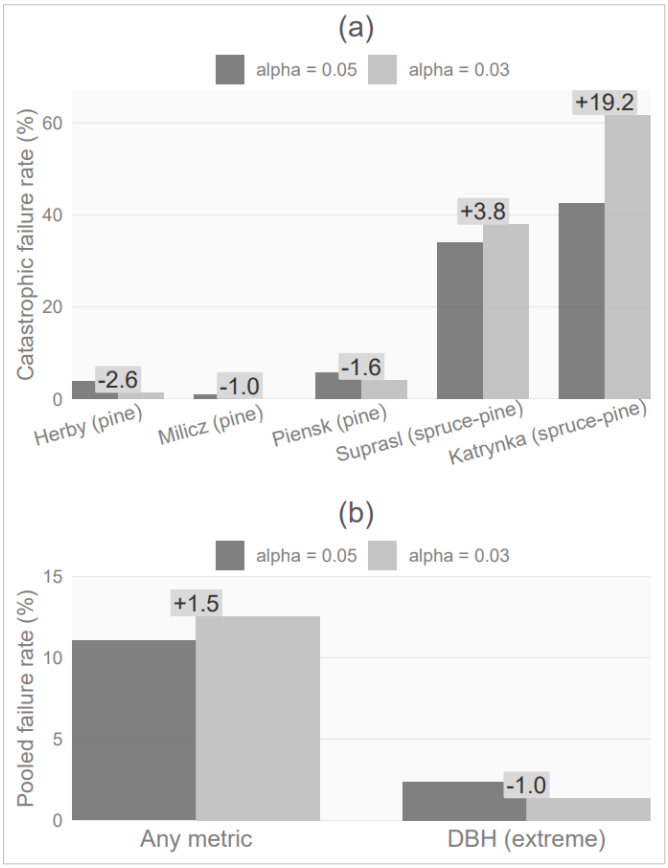


Figure S1. Comparison of catastrophic failure rates between $\alpha = 0.05$ m and $\alpha = 0.03$ m, restricted to bbr = FALSE. **(a)** Per-plot failure rate across the five validation plots. **(b)** Pooled failure rate for any metric and for extreme DBH error. Failure rates are computed from the same F1-filtered reference set used throughout the analysis. Numeric labels above the bars give the change in failure rate relative to $\alpha = 0.05$; positive values indicate higher failure at the finer resolution. The finer mesh produces no consistent improvement: failure rates decrease slightly on Herby, Milicz, and Piensk, insubstantially increase on Suprasl, and substantially increase on Katrynka. Finer mesh resolution therefore does not mitigate the failure modes observed under the default configuration.

9. References

Arrizza, S., Marras, S., Ferrara, R., Pellizzaro, G., 2024. Terrestrial Laser Scanning (TLS) for tree structure studies: a review of methods for wood-leaf classifications from 3D point clouds. Remote Sensing Applications: Society and Environment 36, 101364. https://doi.org/10.1016/j.rsase.2024.101364

Bürkner, P.-C., 2017. brms: An R Package for Bayesian Multilevel Models Using Stan. Journal of Statistical Software 80, 1–28. https://doi.org/10.18637/jss.v080.i01

Burt, A., Disney, M., Calders, K., 2019. Extracting individual trees from lidar point clouds using treeseg. Methods in Ecology and Evolution 10, 438–445. https://doi.org/10.1111/2041-210X.13121

Calders, K., Adams, J., Armston, J., Bartholomeus, H., Bauwens, S., Bentley, L.P., Chave, J., Danson, F.M., Demol, M., Disney, M., Gaulton, R., Krishna Moorthy, S.M., Levick, S.R., Saarinen, N., Schaaf, C., Stovall, A., Terryn, L., Wilkes, P., Verbeeck, H., 2020. Terrestrial laser scanning in forest ecology: Expanding the horizon. Remote Sensing of Environment 251, 112102. https://doi.org/10.1016/j.rse.2020.112102

Calders, K., Newnham, G., Burt, A., Murphy, S., Raumonen, P., Herold, M., Culvenor, D., Avitabile, V., Disney, M., Armston, J., Kaasalainen, M., 2015. Nondestructive estimates of above-ground biomass using terrestrial laser scanning. Methods in Ecology and Evolution 6, 198–208. https://doi.org/10.1111/2041-210X.12301

Carpenter, B., Gelman, A., Hoffman, M.D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., Riddell, A., 2017. Stan: A Probabilistic Programming Language. Journal of Statistical Software 76, 1–32. https://doi.org/10.18637/jss.v076.i01

Chang, L., Fan, H., Zhu, N., Dong, Z., 2022. A Two-Stage Approach for Individual Tree Segmentation From TLS Point Clouds. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 15, 8682–8693. https://doi.org/10.1109/JSTARS.2022.3212445

Cherlet, W., Dayal, K., Chen, S., Cooper, Z., Disney, M., Hanzl, A., Levick, S., Nightingale, J., Origo, N., Senf, C., Soenens, L., Terryn, L., Van den Broeck, W.A.J., Calders, K., 2026. Benchmarking tree instance segmentation of terrestrial laser scanning point clouds. ISPRS Journal of Photogrammetry and Remote Sensing 231, 230–247. https://doi.org/10.1016/j.isprsjprs.2025.10.033

Compeán-Aguirre, J.L., López-Serrano, P.M., Silván-Cárdenas, J.L., Martínez-García-Moreno, C.A., Vega-Nieva, D.J., Corral-Rivas, J.J., Pompa-García, M., 2024. Evaluation of Two-Dimensional DBH Estimation Algorithms Using TLS. Forests 15, 1964. https://doi.org/10.3390/f15111964

Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., Rubin, D.B., 2013. Bayesian Data Analysis, 3rd ed. Chapman and Hall/CRC, New York. https://doi.org/10.1201/b16018

Krisanski, S., Taskhiri, M.S., Gonzalez Aracil, S., Herries, D., Muneri, A., Gurung, M.B., Montgomery, J., Turner, P., 2021. Forest Structural Complexity Tool—An Open Source, Fully-Automated Tool for Measuring Forest Point Clouds. Remote Sensing 13, 4677. https://doi.org/10.3390/rs13224677

Liang, X., Hyyppä, J., Kaartinen, H., Lehtomäki, M., Pyörälä, J., Pfeifer, N., Holopainen, M., Brolly, G., Francesco, P., Hackenberg, J., Huang, H., Jo, H.-W., Katoh, M., Liu, L., Mokroš, M., Morel, J., Olofsson, K., Poveda-Lopez, J., Trochta, J., Wang, D., Wang, J., Xi, Z., Yang, B., Zheng, G., Kankare, V., Luoma, V., Yu, X., Chen, L., Vastaranta, M., Saarinen, N., Wang, Y., 2018. International benchmarking of terrestrial laser scanning approaches for forest inventories. ISPRS Journal of Photogrammetry and Remote Sensing 144, 137–179. https://doi.org/10.1016/j.isprsjprs.2018.06.021

Martin-Ducup, O., Mofack, G., II, Wang, D., Raumonen, P., Ploton, P., Sonké, B., Barbier, N., Couteron, P., Pélissier, R., 2021. Evaluation of automated pipelines for tree and plot metric estimation from TLS data in tropical forest areas. Ann Bot 128, 753–766. https://doi.org/10.1093/aob/mcab051

Peng, X., Yi, J., Liu, R., Shen, X., Li, X., 2026. Individual Tree Segmentation from LiDAR Point Clouds: A Mamba-Enhanced Sparse CNN Approach for Accurate Forest Inventory. Remote Sensing 18, 664. https://doi.org/10.3390/rs18040664

Peucker, T.K., Fowler, R.J., Little, J.J., 1978. Digital representation of three-dimensional surfaces by triangulated irregular networks (TIN). In: Proceedings of the Digital Terrain Modeling Symposium, American Society of Photogrammetry, 516–540.

Roussel, J.-R., Auty, D., Coops, N.C., Tompalski, P., Goodbody, T.R.H., Meador, A.S., Bourdon, J.-F., de Boissieu, F., Achim, A., 2020. lidR: An R package for analysis of Airborne Laser Scanning (ALS) data. Remote Sensing of Environment 251, 112061. https://doi.org/10.1016/j.rse.2020.112061

Shokirov, S., Schaefer, M., Levick, S.R., Jucker, T., Borevitz, J., Abdurahmanov, I., Youngentob, K., 2021. Multi-platform LiDAR approach for detecting coarse woody debris in a landscape with varied ground cover. International Journal of Remote Sensing 42, 9324–9350. https://doi.org/10.1080/01431161.2021.1995072

Stereńczak, K., Kulicki, M., Kraszewski, B., Gasica, T.A., Mitelsztedt, K., Mielcarek, M., Tangwa, E., Erfanifard, Y., Brach, M., Handayani, H.H., Ramadhani, A.N.R., Hawryło, P., Krok, G., Armazeta, A.J.G., Arisandy, D.R., Błaś, F., Damayanti, R.A., De Yong, R.J., Deden, Fabjan, K., Fathimah, F.N., Górska, J., Ila, Q.P.A.N., Irawan, R.D., Iwaniec, F., Jabłoński, A., Krawczyk, W., Kwaśniewski, B., Kwartnik, K., Mardhatillah, R., Massayudevi, R.A., Miszczyszyn, J., Murti, S.B.A., Nabila, D.A., Novatiana, R., Prasetyo, W.E., Pratama, F., Ramadhani, A.T.U., Rębisz, N., Rizannyansyah, H.F.F., Seremak, Z., Skomorucha, B., Skóra, D., Sukmawan, N., Taslyanto, C.A.P., Turemka, G., Valika, E., Warman, D.M., Wilczyńska, A., Winkowska, K., Witkowski, K., Yudisti, S.A., Zdunek, N., Zieliński, D., Ziółkowski, A., 2026. A Central European tree species dataset of annotated terrestrial laser scanning point clouds - TreeScanPL10K. Sci Data. https://doi.org/10.1038/s41597-026-07269-1

Taubin, G., 1991. Estimation of planar curves, surfaces, and nonplanar space curves defined by implicit equations with applications to edge and range image segmentation. IEEE Transactions on Pattern Analysis and Machine Intelligence 13, 1115–1138. https://doi.org/10.1109/34.103273

Thomas, H., Qi, C.R., Deschaud, J.-E., Marcotegui, B., Goulette, F., Guibas, L., 2019. KPConv: Flexible and Deformable Convolution for Point Clouds. 2019 IEEE/CVF International Conference on Computer Vision (ICCV) 6410–6419. https://doi.org/10.1109/ICCV.2019.00651

Trochta, J., Krůček, M., Vrška, T., Král, K., 2017. 3D Forest: An application for descriptions of three-dimensional forest structures using terrestrial LiDAR. PLOS ONE 12, e0176871. https://doi.org/10.1371/journal.pone.0176871

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., Bürkner, P.-C., 2021. Rank-Normalization, Folding, and Localization: An Improved R^ for Assessing Convergence of MCMC (with Discussion). Bayesian Analysis 16, 667–718. https://doi.org/10.1214/20-BA1221

Wang, D., Liang, X., Mofack, G.I., Martin-Ducup, O., 2021. Individual tree extraction from terrestrial laser scanning data via graph pathing. Forest Ecosystems 8, 67. https://doi.org/10.1186/s40663-021-00340-w

Xi, Z., Hopkinson, C., 2021. Detecting Individual-Tree Crown Regions from Terrestrial Laser Scans with an Anchor-Free Deep Learning Model. Canadian Journal of Remote Sensing 47, 228–242. https://doi.org/10.1080/07038992.2020.1861541

Xu, X., Iuricich, F., Calders, K., Armston, J., De Floriani, L., 2023. Topology-based individual tree segmentation for automated processing of terrestrial laser scanning point clouds. International Journal of Applied Earth Observation and Geoinformation 116, 103145. https://doi.org/10.1016/j.jag.2022.103145

Zhang, W., Qi, J., Wan, P., Wang, H., Xie, D., Wang, X., Yan, G., 2016. An Easy-to-Use Airborne LiDAR Data Filtering Method Based on Cloth Simulation. Remote Sensing 8, 501. https://doi.org/10.3390/rs8060501